

21	C	$E = Pt = 40 \times 3 = 120 \text{ J}$
22	A	Characteristics of (1) NTC: when the temperature is high, the resistance of a thermistor decreases. (2) LDR: when light intensity is high, the resistance of the LDR decreases For lamp to glow more brightly, LDR must have lower resistance (to increase the overall supply current) and NTC with a higher resistance (so that more current will pass through lamp in the parallel branch). To achieve this, the light intensity on the LDR must increase while the NTC temperature must decrease.
23	A	The point charge's velocity is parallel to resultant magnetic flux density at the centre of